

Oriented-Support Groupoid and Antipode Quotient

Autonomous discussion Pass 101 packages primitive collapse choices as a signed oriented-support action groupoid.

For finite support S with $|S| \geq 2$, keep the primitive orientation torsor

$$\mathcal{O}_S = \{c = (c_p)_{p \in S} \in \mathbb{Z}^S : \sum_{p \in S} c_p = 0, \gcd(c_p) = 1\}.$$

An object is a pair (S, c) with $c \in \mathcal{O}_S$.

For an inclusion $S \subseteq T$, a morphism $(S, c) \rightarrow (T, d)$ is a sign $\sigma \in \{\pm 1\}$ such that

$$d = \sigma e_{S,T}(c),$$

where $e_{S,T}$ is zero-extension. Identities have sign $+1$, and composition multiplies signs. The antipode is the sign -1 morphism $(S, c) \rightarrow (S, -c)$ over a fixed support, and it squares to the identity.

The coarse antipode quotient

$$[c] = \{c, -c\}$$

presents the primitive collapse line and hence the single all-prime generator, but it loses the signed path label. Therefore the plain quotient does not retain the Pass-94 functional-equation sign. The sign-preserving package is the signed action groupoid itself, or equivalently the coarse quotient equipped with its residual $\mathbb{Z}/2$ sign local system.

On finite N -torsion, the antipode acts by multiplication by -1 on $(\mathbb{Q}/\mathbb{Z})[N]$. This sign is visible for $N > 2$ and collapses at $N = 2$, matching the earlier finite signed-duality behavior.

The result preserves the sign of the shifted constant-term boundary generator without creating a degree-0 Weyl/Fourier map $\epsilon \rightarrow \mathbb{Q}$.

The next task is to push this $\mathbb{Z}/2$ sign local system through $[\mathbb{Q} \rightarrow \mathbb{A}_f]$ and identify the exact boundary or Yoneda class representing biduality on $D\epsilon \simeq \mathbb{Q}[-1]$.

The checker `artifacts/reports/pass101-oriented-support-groupoid-antipode-quotient-check.json` verifies signed morphism closure, multiplicative composition, antipode involutivity, coarse quotient sign loss, sign-local-system restoration, and finite N -torsion sign visibility/collapse.